1 Magnetic Vector Potential, Gauge Freedom, and Uniqueness

We have already introduced the magnetic vector potential $\mathbf{A}$ through

$$\boxed{\mathbf{B} = \nabla \times \mathbf{A}.} \tag{1}$$

This is a very natural definition, because it automatically satisfies one of Maxwell's equations in magnetostatics:

$$\nabla \cdot \mathbf{B} = \nabla \cdot (\nabla \times \mathbf{A}) = 0. \tag{2}$$

Thus, by writing $\mathbf{B}$ as the curl of another vector field, the condition $\nabla \cdot \mathbf{B} = 0$ is built in automatically.

Key idea: Just as in electrostatics we write

$$\mathbf{E} = -\nabla V$$

to automatically satisfy

$$\nabla \times \mathbf{E} = 0,$$

in magnetostatics we write

$$\mathbf{B} = \nabla \times \mathbf{A}$$

to automatically satisfy

$$\nabla \cdot \mathbf{B} = 0.$$

So the scalar potential V is natural for an *irrotational* field, whereas the vector potential $\mathbf{A}$ is natural for a *solenoidal* field.

Electrostatic analogy

Recall the electrostatic case:

$$\nabla \times \mathbf{E} = 0 \quad \Longrightarrow \quad \mathbf{E} = -\nabla V.$$

Then Gauss's law gives

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0} \quad \Longrightarrow \quad \nabla \cdot (-\nabla V) = \frac{\rho}{\varepsilon_0}.$$

Hence,

$$\nabla^2 V = -\frac{\rho}{\varepsilon_0}. \tag{3}$$

So in electrostatics:

- $\nabla \times \mathbf{E} = 0$ motivates introducing V,
- $\nabla \cdot \mathbf{E} = \rho/\varepsilon_0$ determines V,
- together with boundary conditions, V becomes unique up to an additive constant.

Now we ask: what is the corresponding logic in magnetostatics?

Magnetostatic side

Since

$$\nabla \cdot \mathbf{B} = 0,$$

we introduce $\mathbf{A}$ such that

$$\mathbf{B} = \nabla \times \mathbf{A}.$$

Then Ampère's law in magnetostatics is

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J}. \tag{4}$$

Substituting $\mathbf{B} = \nabla \times \mathbf{A}$,

$$\nabla \times (\nabla \times \mathbf{A}) = \mu_0 \mathbf{J}. \tag{5}$$

Now use the vector identity

$$\boxed{\nabla \times (\nabla \times \mathbf{P}) = \nabla(\nabla \cdot \mathbf{P}) - \nabla^2 \mathbf{P}} \tag{6}$$

for any sufficiently smooth vector field $\mathbf{P}$. Therefore,

$$\nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A} = \mu_0 \mathbf{J}. \tag{7}$$

This is the general equation satisfied by the magnetic vector potential.

1.1 Why A Is Not Unique

At first sight, one may think that once $\mathbf{B}$ is known, $\mathbf{A}$ is completely determined from

$$\mathbf{B} = \nabla \times \mathbf{A}.$$

But this is *not* true.

Let us define a new vector field

$$\mathbf{A}' = \mathbf{A} + \nabla \psi, \tag{8}$$

where ψ is any scalar function.

Then

$$\begin{aligned} \nabla \times \mathbf{A}' &= \nabla \times (\mathbf{A} + \nabla\psi) && (9) \\ &= \nabla \times \mathbf{A} + \nabla \times (\nabla\psi). && (10) \end{aligned}$$

But

$$\nabla \times (\nabla\psi) = 0.$$

Hence,

$$\boxed{\nabla \times \mathbf{A}' = \nabla \times \mathbf{A} = \mathbf{B}.} \tag{11}$$

So $\mathbf{A}$ and $\mathbf{A} + \nabla\psi$ produce the *same* magnetic field.

Important conclusion:

$$\boxed{\mathbf{A} \text{ is not unique.}}$$

Any gradient field can be added to $\mathbf{A}$ without changing $\mathbf{B}$, because the curl of a gradient is always zero.

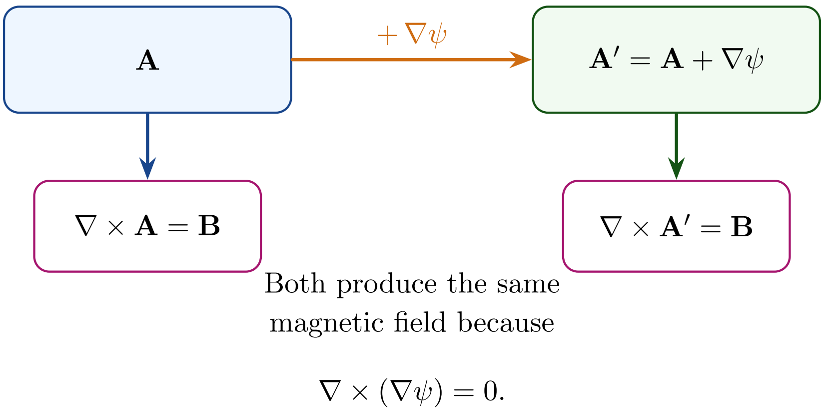


Figure 1: Gauge freedom of the magnetic vector potential. Adding the gradient of any scalar function changes $\mathbf{A}$, but does not change $\mathbf{B} = \nabla \times \mathbf{A}$.

1.2 Using This Freedom: Choice of Gauge

The freedom

$$\mathbf{A} \to \mathbf{A} + \nabla\psi$$

is called a **gauge transformation**.

This freedom is not useless; rather, it is very powerful. It allows us to choose $\mathbf{A}$ in a mathematically convenient form.

Let

$$\mathbf{A}' = \mathbf{A} + \nabla\psi.$$

Then

$$\nabla \cdot \mathbf{A}' = \nabla \cdot \mathbf{A} + \nabla^2\psi. \tag{12}$$

Suppose we want to impose the condition

$$\boxed{\nabla \cdot \mathbf{A}' = 0.} \tag{13}$$

Then from (12), we need

$$\nabla^2\psi = -\nabla \cdot \mathbf{A}. \tag{14}$$

So by solving a Poisson equation for ψ, we can always adjust the divergence of $\mathbf{A}$ to zero (under suitable conditions on the region and boundary behavior).

This choice

$$\nabla \cdot \mathbf{A} = 0$$

is called the **Coulomb gauge**.

Why Coulomb gauge is useful: Once we choose

$$\nabla \cdot \mathbf{A} = 0,$$

the general equation

$$\nabla(\nabla \cdot \mathbf{A}) - \nabla^2\mathbf{A} = \mu_0\mathbf{J}$$

simplifies to

$$\boxed{\nabla^2\mathbf{A} = -\mu_0\mathbf{J}.} \tag{15}$$

Thus each component of $\mathbf{A}$ satisfies a Poisson-type equation, very similar in spirit to the electrostatic potential equation

$$\nabla^2 V = -\frac{\rho}{\varepsilon_0}.$$

1.3 Logical Meaning of Curl and Divergence in Uniqueness

Now let us state the deeper point carefully.

A vector field is not uniquely specified by only one of the two quantities:

- divergence alone is not enough,
- curl alone is not enough.

To determine a vector field uniquely, in general we need:

$$\boxed{\text{divergence} + \text{curl} + \text{boundary conditions.}}$$

For the magnetic field $\mathbf{B}$, these are:

$$\nabla \cdot \mathbf{B} = 0, \qquad \nabla \times \mathbf{B} = \mu_0 \mathbf{J}, \tag{16}$$

together with suitable boundary conditions.

Then $\mathbf{B}$ becomes unique.

But $\mathbf{A}$ itself is still not unique, because many different $\mathbf{A}$'s can produce the same $\mathbf{B}$.

Only after fixing a gauge condition such as

$$\nabla \cdot \mathbf{A} = 0$$

and giving suitable boundary conditions do we select one convenient representative from this family.

Logical summary:

- $\mathbf{B}$ is the physically measurable field.
- $\mathbf{A}$ is an auxiliary potential field introduced to represent $\mathbf{B}$.
- $\mathbf{B}$ can be unique even when $\mathbf{A}$ is not.
- Gauge fixing removes this extra mathematical freedom.

1.4 Comparison with Electrostatics

It is useful to compare the two cases side by side:

Electrostatics	Magnetostatics
$\nabla \times \mathbf{E} = 0$	$\nabla \cdot \mathbf{B} = 0$
$\mathbf{E} = -\nabla V$	$\mathbf{B} = \nabla \times \mathbf{A}$
$\nabla \cdot \mathbf{E} = \rho/\varepsilon_0$	$\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$
$\nabla^2 V = -\rho/\varepsilon_0$	$\nabla^2 \mathbf{A} = -\mu_0 \mathbf{J}$ in Coulomb gauge
V unique up to a constant	$\mathbf{A}$ not unique; gauge freedom exists

Final message:

- The field $\mathbf{B}$ is introduced through $\mathbf{A}$ because $\nabla \cdot \mathbf{B} = 0$.
- The vector potential $\mathbf{A}$ is not unique because adding $\nabla\psi$ does not change $\mathbf{B}$.
- A gauge condition such as $\nabla \cdot \mathbf{A} = 0$ is used to make the equation for $\mathbf{A}$ simpler and more useful.
- A vector field is uniquely determined only when its divergence, curl, and boundary behavior are all specified.

2 Magnetic Field Due to a Small Current Loop at a Far-Away Point

Consider a **small circular current loop** of radius a, centered at the origin and lying in the xy-plane. We wish to find the magnetic field at a **far-away observation point** P, described in spherical coordinates by

$$P \equiv (r, \theta, \phi), \qquad r \gg a.$$

Geometrical meaning of "small loop" and "far-away point":

- The loop is small compared to the observation distance: $a \ll r$.
- The observation point can therefore be described primarily by the large distance r, while the source point $\mathbf{r}'$ moves only over a small region around the origin.
- In cylindrical form, the observation position is

$$\mathbf{r} = \rho\,\hat{\rho} + z\,\hat{\mathbf{z}}, \qquad \rho = r\sin\theta, \qquad z = r\cos\theta.$$

Let a source point on the loop be denoted by $\mathbf{r}'$, and let the corresponding current element be $I\,d\ell'$. The separation vector from the source point to the field point is

$$\mathbf{R} = \mathbf{r} - \mathbf{r}', \qquad R = |\mathbf{R}|.$$

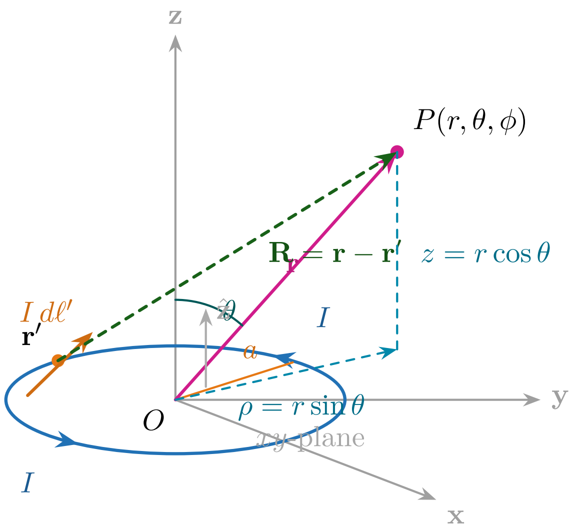


Figure 2: Geometry of a small circular current loop in the xy-plane and a far-away observation point P. The source quantities $\mathbf{r}'$ and $I\,d\ell'$ live on the loop, while the field is evaluated at $\mathbf{r}$.

The magnetic field due to the loop is given by the Biot–Savart law:

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \oint_C \frac{I\,d\ell' \times (\mathbf{r} - \mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|^3}. \tag{17}$$

Why direct use of Biot–Savart is difficult here:

- The direction of $d\ell'$ keeps changing as we move around the loop.
- The vector $(\mathbf{r} - \mathbf{r}')$ also changes from one source point to another.
- Hence the cross product in (17) must be handled point-by-point around the entire loop.

So, although (17) is exact, direct integration is not the most convenient starting point.

A more convenient route is to first compute the **vector potential A**, and then obtain **B** from

$$\mathbf{B} = \nabla \times \mathbf{A}. \tag{18}$$

For a steady current loop,

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \oint_C \frac{I\,d\ell'}{|\mathbf{r} - \mathbf{r}'|}. \tag{19}$$

Why $\mathbf{A}$ is useful:

- In (19), the denominator is still geometry-dependent, but the difficult cross product is gone.
- So the integration becomes cleaner.
- After finding $\mathbf{A}$, the magnetic field follows from $\mathbf{B} = \nabla \times \mathbf{A}$.

2.1 Using Symmetry in the Vector Potential Integral

Without loss of generality, choose the azimuth of the observation point to be zero, so that the point P lies in the ρz-plane. Then the projection of P on the loop plane lies along the $+x$-axis, and at the observation point the cylindrical azimuthal direction is simply

$$\hat{\phi} \equiv \hat{\mathbf{y}}.$$

Key symmetry idea: Consider two current elements on the loop located at source azimuths $+\phi'$ and $-\phi'$. These two elements are symmetrically placed with respect to the xz-plane. Hence they are at the *same distance* from the observation point P. Therefore, in the vector-potential integral, both have the same denominator

$$R = |\mathbf{r} - \mathbf{r}'|.$$

Only the *directions* of their current elements differ.

From the geometry of Fig. 3, the two current elements are

$$d\ell'_A = a\, d\phi' \left(\sin\phi'\, \hat{\mathbf{x}} + \cos\phi'\, \hat{\mathbf{y}}\right), \tag{20}$$

$$d\ell'_B = a\, d\phi' \left(-\sin\phi'\, \hat{\mathbf{x}} + \cos\phi'\, \hat{\mathbf{y}}\right). \tag{21}$$

Now add these two vectors:

$$d\ell'_A + d\ell'_B = a\, d\phi' \left[\left(\sin\phi' - \sin\phi'\right)\hat{\mathbf{x}} + \left(\cos\phi' + \cos\phi'\right)\hat{\mathbf{y}}\right] \tag{22}$$

$$= 2a\cos\phi'\, d\phi'\, \hat{\mathbf{y}}. \tag{23}$$

Thus the components along $\hat{\mathbf{x}}$ cancel, while the components along $\hat{\mathbf{y}}$ add. But at the observation point, $\hat{\mathbf{y}}$ is precisely the cylindrical azimuthal direction $\hat{\phi}$. Hence,

$$d\ell'_A + d\ell'_B = 2a\cos\phi'\, d\phi'\, \hat{\phi}.$$

Therefore, in the vector-potential integral (19), only the $\hat{\phi}$-component survives after summing symmetrically placed source elements. In other words,

$$\mathbf{A} = A_\phi\, \hat{\phi}. \tag{24}$$

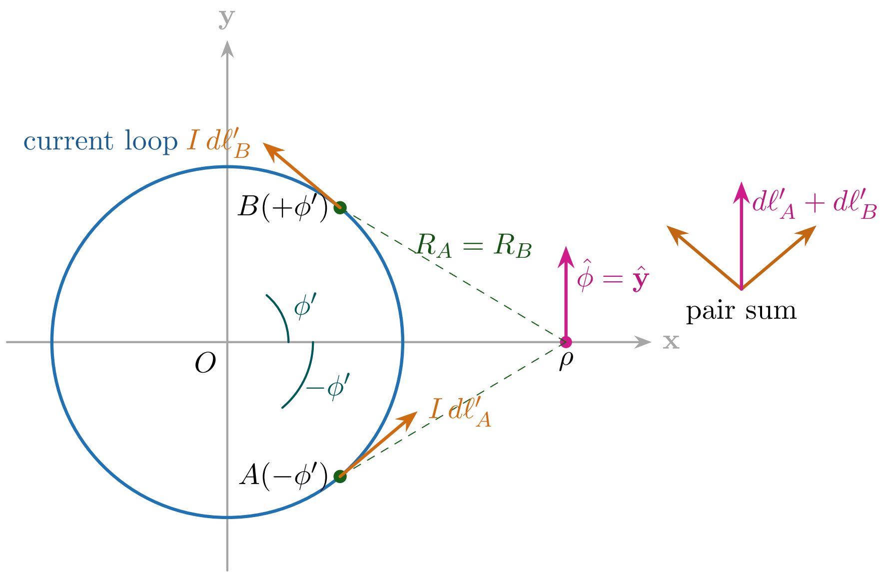


Figure 3: Top view of the circular loop. The two current elements at source azimuths $+\phi'$ and $-\phi'$ are symmetrically placed. Their distances to the observation point are equal, so their denominators are the same. Their horizontal components cancel, while the surviving contribution is along the azimuthal direction $\hat{\phi}$.

So the vector potential can be written as

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0 I}{4\pi} \oint_C \frac{d\ell'}{R} = \frac{\mu_0 I a}{4\pi} \left(\int_0^{2\pi} \frac{\cos\phi'}{R(\phi')}\, d\phi' \right) \hat{\phi}, \tag{25}$$

where

$$R(\phi') = |\mathbf{r} - \mathbf{r}'|.$$

Important conclusion: Because of symmetry, the vector potential of the circular loop has only an $\hat{\phi}$-component:

$$\boxed{\mathbf{A} = A_\phi\, \hat{\phi}}.$$

Thus the difficult vector nature of the integral is already simplified. The next step is only to evaluate the scalar denominator $R(\phi') = |\mathbf{r} - \mathbf{r}'|$.

2.2 Evaluating the Denominator $R(\phi') = |\mathbf{r} - \mathbf{r}'|$

We now evaluate the quantity $R(\phi')$ appearing in (25). This is the geometric heart of the derivation.

Recall that the observation point is chosen in the xz-plane, so that

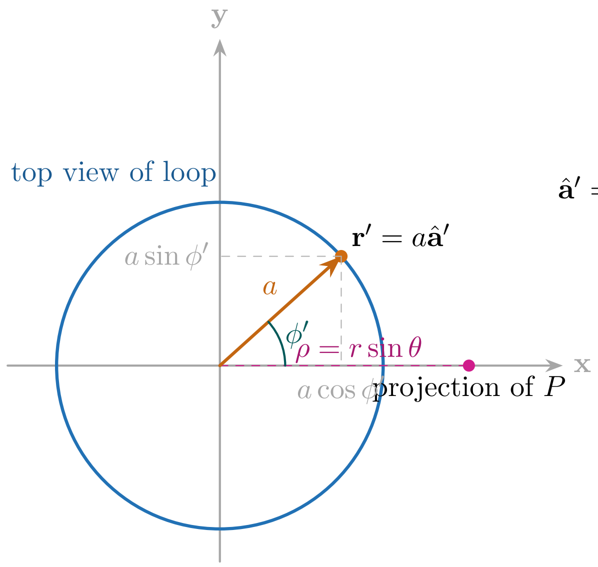
$$\mathbf{r} = \rho\,\hat{\mathbf{x}} + z\,\hat{\mathbf{z}} = r\sin\theta\,\hat{\mathbf{x}} + r\cos\theta\,\hat{\mathbf{z}}.$$

A source point on the loop of radius a is

$$\mathbf{r}' = a(\cos\phi'\,\hat{\mathbf{x}} + \sin\phi'\,\hat{\mathbf{y}}).$$

Thus the denominator is

$$R(\phi') = |\mathbf{r} - \mathbf{r}'|.$$



$$\hat{\mathbf{a}}' = \cos\phi'\,\hat{\mathbf{x}} + \sin\phi'\,\hat{\mathbf{y}}$$

Since the observer lies in the xz-plane, its projection on the loop plane is along $+\hat{\mathbf{x}}$. Hence the relevant projection of $\hat{\mathbf{a}}'$ along $\hat{\mathbf{r}}$ comes through $\cos\phi'$.

Figure 4: Top view of the loop plane. The source point is at azimuth ϕ'. Since the observation point lies in the xz-plane, the projection of $\mathbf{r}$ on the loop plane is along $+\hat{\mathbf{x}}$.

Step 1: Use the law of cosines

Let ψ be the angle between $\mathbf{r}$ and the loop radius direction $\hat{\mathbf{a}}'$. Then, by the law of cosines,

$$R^2 = |\mathbf{r} - \mathbf{r}'|^2 = r^2 + a^2 - 2ar\cos\psi. \tag{26}$$

So the problem reduces to finding $\cos\psi$.

Step 2: Compute $\cos\psi$ from a dot product

By definition,

$$\cos\psi = \hat{\mathbf{r}} \cdot \hat{\mathbf{a}}'.$$

Now,

$$\hat{\mathbf{r}} = \sin\theta\,\hat{\mathbf{x}} + \cos\theta\,\hat{\mathbf{z}},$$

because the observer is in the xz-plane, and

$$\hat{\mathbf{a}}' = \cos\phi'\,\hat{\mathbf{x}} + \sin\phi'\,\hat{\mathbf{y}}.$$

Therefore,

$$\cos\psi = (\sin\theta\,\hat{\mathbf{x}} + \cos\theta\,\hat{\mathbf{z}}) \cdot (\cos\phi'\,\hat{\mathbf{x}} + \sin\phi'\,\hat{\mathbf{y}}) \tag{27}$$

$$= \sin\theta\cos\phi'. \tag{28}$$

Important geometric meaning: Only the part of the loop radius direction lying along the observer's projection on the loop plane contributes to the dot product. That is why $\cos\psi$ becomes

$$\cos\psi = \sin\theta\cos\phi'.$$

Substituting this into (26), we get

$$R^2 = r^2 + a^2 - 2ar\sin\theta\cos\phi'. \tag{29}$$

Hence,

$$R = \left(r^2 + a^2 - 2ar\sin\theta\cos\phi'\right)^{1/2}. \tag{30}$$

Now substitute (30) into the vector-potential expression (25):

$$\mathbf{A} = \hat{\phi}\,\frac{\mu_0 I a}{4\pi}\int_0^{2\pi} \frac{\cos\phi'\,d\phi'}{(r^2 + a^2 - 2ar\sin\theta\cos\phi')^{1/2}}. \tag{31}$$

2.3 Far-Field Approximation: $r \gg a$

We now use the fact that the loop is *small* and the observation point is *far away*, namely

$$a \ll r.$$

Start from (30):

$$R = r\left(1 + \frac{a^2}{r^2} - 2\frac{a}{r}\sin\theta\cos\phi'\right)^{1/2}.$$

Therefore,

$$\frac{1}{R} = \frac{1}{r}\left(1 + \frac{a^2}{r^2} - 2\frac{a}{r}\sin\theta\cos\phi'\right)^{-1/2}. \tag{32}$$

Since $a/r \ll 1$, keep terms only up to first order in a/r. Using the binomial approximation

$$(1+u)^{-1/2} \approx 1 - \frac{u}{2}, \qquad |u| \ll 1,$$

with

$$u = \frac{a^2}{r^2} - 2\frac{a}{r}\sin\theta\cos\phi',$$

and neglecting the second-order term a^2/r^2, we obtain

$$\frac{1}{R} \approx \frac{1}{r}\left(1 + \frac{a}{r}\sin\theta\cos\phi'\right). \tag{33}$$

Why the first term alone is not enough: If we keep only $1/R \approx 1/r$, then in (31) the integral becomes proportional to

$$\int_0^{2\pi} \cos\phi'\, d\phi' = 0.$$

So the leading nonzero contribution comes from the *next term*, namely

$$\frac{a}{r}\sin\theta\cos\phi'.$$

This is why we must retain the first-order correction in a/r.

Substitute (33) into (31):

$$\mathbf{A} \approx \hat{\phi}\,\frac{\mu_0 I a}{4\pi r}\int_0^{2\pi}\cos\phi'\left(1 + \frac{a}{r}\sin\theta\cos\phi'\right)d\phi' \tag{34}$$

$$= \hat{\phi}\,\frac{\mu_0 I a}{4\pi r}\left[\int_0^{2\pi}\cos\phi'\,d\phi' + \frac{a}{r}\sin\theta\int_0^{2\pi}\cos^2\phi'\,d\phi'\right]. \tag{35}$$

Now use the standard integrals

$$\int_0^{2\pi}\cos\phi'\,d\phi' = 0, \qquad \int_0^{2\pi}\cos^2\phi'\,d\phi' = \pi.$$

Hence (35) becomes

$$\mathbf{A} = \hat{\phi}\,\frac{\mu_0 I a}{4\pi r}\left(\frac{a}{r}\sin\theta\right)\pi. \tag{36}$$

So finally,

$$\boxed{\mathbf{A} = \hat{\phi}\,\frac{\mu_0 I a^2}{4r^2}\sin\theta} \tag{37}$$

2.4 Introducing the Magnetic Moment

The area of the circular loop is

$$S = \pi a^2.$$

So (37) can be written as

$$\mathbf{A} = \hat{\phi}\,\frac{\mu_0}{4\pi}\,\frac{(I\pi a^2)\sin\theta}{r^2}. \tag{38}$$

This naturally suggests defining the **magnetic moment** of the loop as

$$\boxed{\mathbf{m} = I\,S\,\hat{\mathbf{n}}} \tag{39}$$

where:

- I is the loop current,
- S is the loop area,
- $\hat{\mathbf{n}}$ is the unit normal to the loop, given by the right-hand rule.

For the present loop in the xy-plane,

$$\hat{\mathbf{n}} = \hat{\mathbf{z}}, \qquad \mathbf{m} = I\pi a^2\,\hat{\mathbf{z}}.$$

Therefore (38) becomes

$$\boxed{\mathbf{A} = \hat{\phi}\,\frac{\mu_0}{4\pi}\,\frac{m\sin\theta}{r^2}} \tag{40}$$

where $m = |\mathbf{m}|$.

Insight: what does magnetic moment really mean?
The magnetic moment is the *single quantity* that summarizes the magnetic strength and orientation of a small current loop.

- A larger current I means charges circulate more strongly $\Rightarrow$ stronger magnetic effect.
- A larger area S means the current loop encloses more area $\Rightarrow$ stronger dipole action.
- The direction $\hat{\mathbf{n}}$ tells the *axis* of the tiny magnet formed by the loop.

So,

$$\boxed{\text{magnetic moment} = \text{strength of the loop as a tiny magnet}}$$

together with the direction in which that tiny magnet points.
This is why a small current loop behaves, at far distances, like a *magnetic dipole*.

Right-hand-rule picture: Curl the fingers of your right hand in the direction of current around the loop. Your thumb then points along $\mathbf{m}$. So $\mathbf{m}$ is not along the wire; it is perpendicular to the plane of the loop.

We can write (40) in a compact vector form. Since $\mathbf{m} = m\,\hat{\mathbf{z}}$, one may verify that

$$\mathbf{m} \times \hat{\mathbf{r}} = m \sin\theta\,\hat{\phi}.$$

Hence

$$\boxed{\mathbf{A} = \frac{\mu_0}{4\pi}\frac{\mathbf{m} \times \hat{\mathbf{r}}}{r^2}} \tag{41}$$

or equivalently,

$$\boxed{\mathbf{A} = \frac{\mu_0}{4\pi}\frac{\mathbf{m} \times \mathbf{r}}{r^3}.} \tag{42}$$

Final far-field result for vector potential of a small loop:

$$\boxed{\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi}\frac{\mathbf{m} \times \mathbf{r}}{r^3}}$$

This has exactly the same structure as the vector potential of a magnetic dipole. Thus, at far distances, the small current loop is fully characterized by its magnetic moment $\mathbf{m}$.

What comes next? The next step is to compute

$$\mathbf{B} = \nabla \times \mathbf{A}$$

using (42), which gives the familiar far-field magnetic dipole field of a small current loop.